Python Code

```python
circle = [(5,1), (2,2), (1,5), (2,8), (5,9), (8,8), (9,5), (8,2)]

def pattern_circle():
    print('CIRCLE')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in circle:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_circle()


Rhombus = [(1,3), (1,4), (1,5), (1,6), (1,7), (3,2),
           (3,6), (5,1), (5,2), (5,3), (5,4), (5,5)]

def pattern_Rhombus():
    print('RHOMBUS')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in Rhombus:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_Rhombus()


oval = [(3,4), (3,5), (3,6), (4,2), (4,8),
        (5,1), (5,9), (6,1), (6,9), (7,2), (7,8), (8,4),
        (8,5), (8,6)]

def pattern_oval():
    print('OVAL')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in oval:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_oval()


Triangle = [(2,5), (3,4), (3,6), (4,3), (4,7), (5,2),
            (5,3), (5,4), (5,5), (5,6), (5,7), (5,8)]
```

```
def pattern_Triangle():
    print('TRIANGLE')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in Triangle:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_Triangle()


square = [(2,2), (2,3), (2,4), (2,5), (2,6), (2,7),
          (2,8), (3,2), (3,8), (4,2), (4,8), (5,2),
          (5,8), (6,2), (6,8), (7,2), (7,8), (8,2),
          (8,3), (8,4), (8,5), (8,6), (8,7), (8,8)]

def pattern_square():
    print('SQUARE')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in square:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_square()


Semi_circle = [(4,4), (4,5), (4,6), (4,7), (5,3),
               (5,8), (6,2), (6,9), (7,2), (7,9),
               (8,2), (8,3), (8,4), (8,5), (8,6),
               (8,7), (8,8), (8,9)]

def pattern_SEMI_CIRCLE():
    print('SEMI CIRCLE')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in Semi_circle:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_SEMI_CIRCLE()


rec = [(1,2), (1,3), (1,4), (1,5), (1,6),
       (1,7), (1,8), (1,9), (2,2), (2,9),
       (3,2), (3,9), (4,2), (4,3), (4,4),
       (4,5), (4,6), (4,7), (4,8), (4,9)]
```

```python

def pattern_rectangle():
    print('RECTANGLE')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in rec:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_rectangle()


hexagon = [(2,4), (2,5), (2,6), (3,3), (3,7),
           (4,2), (4,8), (5,1), (5,9), (6,2), (6,8),
           (7,3), (7,7), (8,4), (8,5), (8,6)]

def pattern_hex():
    print('HEXAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in hexagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_hex()


trapezoid = [(2,3), (2,4), (2,5), (2,6), (2,7),
             (3,2), (3,8), (4,1), (4,9), (5,1),
             (5,2), (5,3), (5,4), (5,5), (5,6),
             (5,7), (5,8), (5,9)]

def pattern_trap():
    print('TRAPEZOID')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in trapezoid:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_trap()


heptagon = [(1,5), (2,3), (2,7), (3,2), (3,8),
            (4,2), (4,8), (5,2), (5,8), (6,3),
            (6,7), (7,4), (7,5), (7,6)]

def pattern_hep():
```

```python
    print('HEPTAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in heptagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_hep()


pentagon = [(1,5), (3,3), (3,7), (4,2), (4,8), (6,3), (6,7), (7,4), (7,5), (7,6)]

def pattern_pen():
    print('PENTAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in pentagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_pen()


octagon = [(2,4), (2,5), (2,6), (3,3), (3,7), (4,2), (4,8), (5,2), (5,8), (6,2), (6,8), (7,3), (7,7), (8,4), (8,5), (8,6)]

def pattern_oct():
    print('OCTAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in octagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_oct()


heart = [(2,2), (2,3), (2,7), (2,8), (3,1), (3,4), (3,6), (3,9), (4,1), (4,9), (5,2), (5,8), (6,3), (6,7), (7,4), (7,6), (8,5)]

def pattern_heart():
    print('HEART')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in heart:
                print('*', end=' ')
            else:
                print(' ', end=' ')
```

```
        print()

pattern_heart()


nonagon = [(1,5), (2,3), (2,7), (4,1), (4,9), (6,1), (6,9), (8,3), (8,7), (9,4), (9,5), (9,6)]

def pattern_non():
    print('NONAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in nonagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_non()

decagon = [(1,4), (1,5), (1,6), (2,3), (2,7), (3,2), (3,8), (4,2), (4,8), (5,2), (5,8), (6,2), (6,8), (7,3), (7,7), (8,4), (8,5), (8,6)]

def pattern_dec():
    print('DECAGON')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in decagon:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_dec()


star = [(1,5), (2,3), (2,6), (3,2), (3,8), (4,3), (4,7), (5,2), (5,5), (5,8), (6,1), (6,3), (6,7), (6,9)]

def pattern_star():
    print('STAR')
    for i in range(1, 10):
        for j in range(1, 10):
            if (i, j) in star:
                print('*', end=' ')
            else:
                print(' ', end=' ')
        print()

pattern_star()



smiley = [(3,3), (3,7),(6,2), (7,3), (8,4), (8,5), (8,6), (7,7), (6,8)]
```

```
def pattern_smiley():
    print('SMILEY FACE')
    for i in range(1, 11):
        for j in range(1, 11):
            if (i, j) in smiley:
                print('+', end=' ')
            else:

                if (i==1 and j>3 and j<8) or (i==10 and j>3 and j<8) or \
                   (j==1 and i>3 and i<8) or (j==10 and i>3 and i<8) or \
                   (i==2 and (j==2 or j==9)) or (i==9 and (j==2 or j==9)):
                    print('.', end=' ')
                else:
                    print(' ', end=' ')
        print()

pattern_smiley()



heart_outer = [
    (2,4), (2,5), (2,8), (2,9),
    (3,3), (3,6), (3,7), (3,10),
    (4,2), (4,11),
    (5,2), (5,11),
    (6,3), (6,10),
    (7,4), (7,9),
    (8,5), (8,8),
    (9,6), (9,7)
]

heart_inner = [
    (3,4), (3,5), (3,8), (3,9),
    (4,3), (4,6), (4,7), (4,10),
    (5,3), (5,10),
    (6,4), (6,9),
    (7,5), (7,8),
    (8,6), (8,7)
]

def pattern_double_heart():
    print('\nDOUBLE LAYER HEART')
    for i in range(1, 13):
        for j in range(1, 13):

            if (i, j) in heart_inner:
                print('*', end=' ')
            elif (i, j) in heart_outer:
                print('+', end=' ')
            else:
                print(' ', end=' ')
        print()


pattern_double_heart()
```

```
>>>
================================ RESTART: C:/Python314/GEN-AI-VL-A6.3.py =======
CIRCLE
        *
  *           *


*               *


  *           *
        *
RHOMBUS
    * * * * *

  *       *

* * * * *
```

OVAL

```
      * * *
  *           *
*               *
*               *
  *           *
      * * *
```

TRIANGLE

```
        *
      *   *
    *       *
  * * * * * * *
```

SQUARE

```
* * * * * * *
*           *
*           *
*           *
*           *
*           *
* * * * * * *
```

SEMI CIRCLE

```
    * * * *
  *         *
*             *
*             *
* * * * * * * *
```

RECTANGLE

```
* * * * * * * *
*             *
*             *
* * * * * * * *
```

HEXAGON

```
      * * *
    *       *
  *           *
*               *
  *           *
    *       *
      * * *
```

TRAPEZOID

```
    * * * * *
  *           *
*               *
* * * * * * * * *
```

HEPTAGON

```
      *
  *       *
*           *
*           *
*           *
  *       *
    * * *
```

PENTAGON

```
      *

  *       *
*           *

  *       *
    * * *
```

OCTAGON

```
      * * *
    *       *
  *           *
  *           *
  *           *
    *       *
      * * *
```

HEART

```
  * *       * *
*     *   *     *
*               *
  *           *
    *       *
      *   *
        *
```

NONAGON

```
    *
  *   *

*       *

*       *

  *   *
   ***
```

DECAGON

```
   ***
  *   *
 *     *
 *     *
 *     *
 *     *
  *   *
   ***
```

STAR

```
        *
    *     *
  *           *
    *       *
  *     *     *
*   *       *   *
```

SMILEY FACE

```
      . . . .
  .             .
    +       +
.                 .
.                 .
. +           +   .
.   +       +     .
      + + +
  .             .
      . . . .
```

DOUBLE LAYER HEART

```
    + +     + +
  + * * + + * * +
+ *     * *     * +
+ *             * +
  + *         * +
    + *     * +
      + * * +
        + +
```